

Generation Interconnection System Impact Study Report

For Queue Project Q20-1586

Twin Branch Pv 100 kV

December 17, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1586. The Interconnection Customer proposes a 157.1 MW solar photovoltaic facility interconnecting at the Twin Branch Pv 100 kV substation in the Indiana area. The request was received on October 06, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 544 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$3,721,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
HIGH FALLS - CROGHAN NY-SCENIC WOODS 100.0 kV	PARKROSE-TWIN BRANCH PV 100.0 kV	125.2	223	100.0
PARKROSE-TWIN BRANCH PV 100.0 kV L	SOUTH BEND-HIGH FALLS - CROGHAN NY 100.0 kV Line	125.1	248	64.8

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
SCENIC WOODS-HIGH FALLS - CROGHAN NY 100.0 kV	Rebuild line with higher-capacity conductor	\$2,651,000	70% (\$1,860,000)
PARKROSE-TWIN BRANCH PV 100.0 kV L	Rebuild line with higher-capacity conductor	\$6,001,000	31% (\$1,861,000)
	Total allocated to project		\$3,721,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.